

Project Proposal Form

Go-Around Classification Using ADS-B and METAR Data

1 Team Members and Task Distribution

Member	Planned Responsibilities
Furkan Güney Student ID: 704241023	Literature review, project framing, backend API, Docker packaging, HTML client interface, report writing, and presentation preparation.
Alper Berkin Yazıcı Student ID: 704241020	Data preprocessing, ADS-B and METAR feature engineering, model implementation/training, hyperparameter tuning, evaluation, and error analysis.
Joint Work	Final problem formulation, model selection, interpretation of results, live demo, and final presentation.

2 Problem Definition

A go-around (missed approach) is an aborted landing during final approach or immediately after touchdown. Although it is a safety-preserving maneuver, it increases controller and pilot workload, reduces runway efficiency, and may propagate delays. Predicting go-arounds before touchdown is therefore an important aviation pattern recognition problem and is closely related to data-driven operational decision support in avionics and air traffic operations.

We formulate the task as binary classification. For each arrival, let

$$y_i \in \{0, 1\},$$

where $y_i = 1$ denotes a go-around and $y_i = 0$ denotes a successful landing. Each flight is represented by a feature vector

$$\mathbf{x}_i \in \mathbb{R}^d$$

constructed from ADS-B trajectory information and METAR weather observations during final approach. The goal is to learn a decision function

$$f(\mathbf{x}_i) \rightarrow y_i.$$

This study aims to investigate whether public ADS-B and METAR data can be used to predict go-around risk with satisfactory classification performance. The proposed study is aligned with the course scope, as it involves supervised learning, classification, probabilistic decision making, and empirical evaluation.

3 Dataset Description

The primary data source will be the publicly available *Large Landing Trajectory Dataset for Go-Around Analysis*, derived from OpenSky Network observations. We plan to use the `go_around augmented.csv.gz` file because it already includes landing labels together with airport, runway, aircraft, and METAR-related attributes. The full dataset contains almost 9 million landings, more than 33,000 go-arounds, 176 airports, and 44 countries for 2019, which makes it suitable for both realistic modeling and controlled subsetting.

Planned feature groups are:

- **ADS-B/trajectory features:** altitude, vertical rate, ground speed, heading/track change, distance to runway threshold, runway alignment, and rolling statistics from the final approach segment.
- **Operational/context features:** airport/runway, aircraft type category, wake turbulence category, runway geometry, traffic density proxy, and time-of-day.
- **METAR features:** wind speed/direction, gust, visibility, pressure, temperature, and present weather indicators/codes.

Because go-arounds are rare, the dataset is imbalanced. We will therefore use class-aware training and evaluation metrics beyond accuracy. To keep the project manageable, the initial experiments will focus on a selected set of airports with sufficient go-around samples; extension to a broader multi-airport setting will be considered subsequently. Potential training and calibration trajectories will also be filtered, where necessary, using dataset metadata such as the number of approaches.

4 Proposed Methodology

Preprocessing. We will clean incomplete records, align variables to a consistent observation window, encode categorical variables, and standardize numerical features when needed. Data splits will be designed to avoid leakage across highly similar samples; we will prefer airport-aware and/or time-aware splitting strategies depending on the final subset.

Models. To align the project with core course topics, we plan to compare at least three classifiers:

1. Linear Discriminant Analysis (LDA) as a classical pattern recognition baseline,
2. one tree-based model (Decision Tree / Random Forest / Gradient Boosting),
3. a Multi-Layer Perceptron (MLP) as a neural baseline.

In addition, Logistic Regression will be included as an auxiliary probabilistic baseline.

For a linear probabilistic baseline,

$$p(y_i = 1 \mid \mathbf{x}_i) = \sigma(\mathbf{w}^T \mathbf{x}_i + b),$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$. The predicted label is

$$\hat{y}_i = \begin{cases} 1, & p(y_i = 1 \mid \mathbf{x}_i) \geq \tau, \\ 0, & \text{otherwise.} \end{cases}$$

Training can be formulated with binary cross-entropy:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)].$$

Imbalance handling and ablation. Since go-arounds are rare, class weighting and/or resampling strategies will be investigated. To justify the project title and quantify the value of weather information, an ablation study is planned to compare **ADS-B only** features against **ADS-B + METAR** features.

System and Deployment Plan

To satisfy the course deployment requirements, the final trained classifier will be packaged in a single Dockerized application. A lightweight backend API, such as FastAPI, will expose a prediction endpoint that accepts either manual feature input or a small file-based request derived from the selected feature schema. A simple HTML-based web interface within the same container will allow the user to submit input features and receive the predicted class together with a confidence or probability score. This design is intended to ensure reproducibility, support in-class demonstration, and remain fully aligned with the course requirement of serving the model through an API and a minimal web front-end.

5 Literature Review

Prior work shows that go-arounds can be detected and analyzed from open ADS-B data. Proud presented an ADS-B-based go-around detection method and demonstrated that large-scale detection is feasible from crowd-sourced trajectory data. Figuet et al. showed that open ADS-B and meteorological data can be used for go-around prediction, while Monstein et al. later introduced a much larger public landing dataset that supports broader multi-airport experiments.

Kumar et al. studied go-arounds as a classification problem in commercial aviation using ADS-B data. Dhief et al. proposed a machine-learned go-around probability model for final approach and emphasized the difficulty of prediction under class imbalance and limited separability. More recently, Liu et al. studied real-time go-around prediction at JFK with sequence modeling and an operational web-based demonstration.

The proposed project builds on this line of work by using a public large-scale dataset with METAR attributes, by explicitly evaluating the contribution of weather features through an ADS-B-only vs. ADS-B+METAR comparison, and by comparing classical, tree-based, and neural classifiers within a reproducible deployment-oriented pipeline.

6 Evaluation Plan

We will evaluate the problem as binary classification. Since go-arounds are rare, accuracy alone is insufficient. Planned metrics are:

- Accuracy,
- Precision,
- Recall,
- F1-score,
- ROC-AUC,
- PR-AUC,
- confusion matrix.

We will use train/validation/test splits or cross-validation depending on the final subset size, with stratification when appropriate. Hyperparameters will be selected only on the validation data. In addition to comparing models, we will compare ADS-B-only and ADS-B+METAR feature sets, inspect false positives and false negatives, and analyze feature importance or airport/weather-specific error patterns as a secondary analysis.

7 Timeline

Period	Planned Work
08–15 Mar 2026	Finalize proposal, confirm data source/file choice, set project scope, and prepare repository structure.
16–29 Mar 2026	Preprocess the augmented dataset, inspect class balance, define the feature schema, and prepare train/validation/test splits.
30 Mar–12 Apr 2026	Implement LDA and tree-based baselines; run initial experiments.
13–26 Apr 2026	Implement MLP baseline, perform tuning, and run ADS-B-only vs. ADS-B+METAR ablation experiments.
27 Apr–03 May 2026	Build backend API, Dockerize the project, and connect the HTML interface.
04–10 May 2026	Finalize report, prepare slides, rehearse live demo, and perform reproducibility checks.

References

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